EXP. NO: 5

DEVELOP A CUSTOMIZED PING COMMAND TO TEST THE SERVER CONNECTIVITY

AIM:

To develop a customized ping command in python that checks the connectivity between a client and a Server by Sending ICMP Echo Request packets and displaying the response time and Status.

STEPS:

1. Install Python and VS code
2. Create a python file named ping.py
3. Import the required modules
4. Accept the hostname or IP address
5. Detect the operating System
6. use ping -n 4 on windows or ping -c 4 on Linux / mac OS
7. Execute the ping command
8. Display ping output
9. Display Reachable or unreachable

RESULT:

Thus, a ping command in python that checks connectivity between client and Server and Displayed

```python
import subprocess
import platform
print("="*50)
print("Customized Ping Utility")
print("="*50)
host = input("Enter Website or IP Address: ")
system = platform.system()
if system == "Windows":
    command = ["ping", "-n", "4", host]
else:
    command = ["ping", "-c", "4", host]
try:
    result = subprocess.run(command, capture_output=True, text=True)
    print("\nPing Results\n")
    print(result.stdout)
    if result.returncode == 0:
        print("Server Status : Reachable")
    else:
        print("Server Status : Unreachable")
except Exception as e:
    print("Error:", e)
```

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

PS D:\Computer Network> d:; cd 'd:\Computer Network'; & 'c:\Users\REG\AppData\Local\Pro
in32-x64\bundled\libs\debugpy\launcher' '54199' '--' 'd:\Computer Network\g.py'
==================================================
Customized Ping Utility
==================================================
Enter Website or IP Address: google.com

Ping Results


Pinging google.com [142.251.221.142] with 32 bytes of data:
Reply from 142.251.221.142: bytes=32 time=29ms TTL=120
Reply from 142.251.221.142: bytes=32 time=28ms TTL=120
Reply from 142.251.221.142: bytes=32 time=53ms TTL=120
Reply from 142.251.221.142: bytes=32 time=28ms TTL=120

Ping statistics for 142.251.221.142:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 28ms, Maximum = 53ms, Average = 32ms

Server Status : Reachable
```